Lecture 7: Interest-Rate Models

Table of contents

Learning Objectives . . . 2

1 Why Do We Use Mathematical Models? 2
1.1 Mathematical Modeling in Fixed Income . . . 3
1.2 Mathematical Modeling of Interest Rates . . . 3
1.3 What you should remember . . . 4

2 Mathematical Description of One-Factor Interest-Rate Models 4
A basic continuous-time stochastic process . . . 4
Itô process . . . 5
Specifying the dynamics of the drift term . . . 5
Specifying the dynamics of the volatility term . . . 7

3 Arbitrage-Free versus Equilibrium Models 7
Arbitrage-free models . . . 9
Equilibrium models . . . 9

4 When Do We Need a No-Arbitrage Short-Rate Model? 10
4.1 Yield curve versus short-rate model . . . 11
4.2 From a short-rate process to a yield curve . . . 11
4.3 A simple short-rate tree intuition . . . 12
4.4 The Hull–White model . . . 13
4.5 Calibration logic . . . 14

5 Empirical Evidence on Interest-Rate Changes 15
Volatility of rates and the level of interest rates . . . 16
Negative interest rates . . . 16

6 Selecting an Interest-Rate Model 16

7 Estimating Interest-Rate Volatility Using Historical Data **17**
Historical volatility 17
Implied volatility 18
Key Points 18
Suggested Readings 19
Real-Market Data and Resources 19

Learning Objectives

After completing this lecture you will be able to:

- Define an interest-rate model and interpret its key components (drift, volatility, and mean reversion).
- Explain the mathematics behind one-factor stochastic processes used to model interest rates.
- Distinguish between **arbitrage-free** and **equilibrium** interest-rate models and understand the purpose of each.
- Explain when a stochastic short-rate model is needed, and when a discount curve alone is sufficient.
- Describe how the Hull–White model fits today's yield curve while modeling future rate dynamics.
- Discuss empirical evidence on interest-rate changes, including how volatility relates to rate levels and the possibility of negative rates.
- Describe criteria for selecting an interest-rate model in practice.
- Compute historical interest-rate volatility using real or simulated data.

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1 Why Do We Use Mathematical Models?

A **mathematical model** is a simplified representation of reality. It takes a complex situation and describes the most important relationships using equations, assumptions, and parameters.

Models are not exact copies of the real world. They are tools for organizing thinking, making predictions, valuing securities, and comparing risks. A useful model does not need to capture everything. It needs to capture the features that matter for the decision being made.

For example, a bond pricing model may ignore many details about the economy but still be useful if it correctly discounts promised cash flows. A risk model may not predict the exact future path of interest rates, but it can still show how sensitive a portfolio is to rate changes.

Model intuition

A model is useful when it helps answer a specific question better than intuition alone. The right model depends on the question.

1.1 Mathematical Modeling in Fixed Income

In fixed income, models are used for several related but different tasks:

- valuing bonds and derivatives;
- estimating risk exposures such as duration, convexity, and spread sensitivity;
- simulating future interest-rate scenarios;
- pricing embedded options;
- stress testing portfolios;
- comparing relative value across securities.

The simplest fixed-income model is discounted cash flow valuation:

$$P = \sum_i CF_i \times P(0, t_i).$$

This model says that a bond's price equals the present value of its future cash flows. For a plain fixed-rate bond, this may be enough. If the cash flows are known and the discount curve is available, we do not need a complicated model of future interest rates.

But many fixed-income securities are more complicated. Callable bonds, mortgage-backed securities, caps, floors, swaptions, and structured notes have values that depend on what interest rates do in the future. For these instruments, today's yield curve is not enough. We need a model that describes how future interest rates may evolve.

1.2 Mathematical Modeling of Interest Rates

An **interest-rate model** describes the behavior of interest rates over time. It usually starts by choosing one or more state variables. In a one-factor short-rate model, the state variable is the short rate, denoted r_t.

The model then specifies how r_t changes through time. A simple version is:

$$\mathrm{d}r_t = b\,\mathrm{d}t + \sigma\,\mathrm{d}z_t.$$

This equation separates interest-rate movements into two parts:

- the **drift** term, $b\,\mathrm{d}t$, which represents the expected direction of movement;
- the **random shock** term, $\sigma\,\mathrm{d}z_t$, which represents unexpected changes.

The purpose is not to predict the exact future short rate. The purpose is to generate possible future paths of interest rates in a disciplined way.

Those paths can then be used to answer questions such as:

- What might the yield curve look like one year from now?
- How much could a bond portfolio gain or lose if rates move?
- When would an issuer call a callable bond?
- What is the fair value of an interest-rate option?
- How does interest-rate volatility affect the value of a security?

1.3 What you should remember

A yield curve is a snapshot of current market prices. An interest-rate model is a way to generate possible future yield curves.

For plain bonds, the current discount curve is often sufficient. For securities whose cash flows depend on future rates, a model of interest-rate dynamics becomes necessary.

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2 Mathematical Description of One-Factor Interest-Rate Models

An **interest-rate model** is a probabilistic description of how interest rates evolve over time. In fixed-income portfolio management, models help value bonds and derivatives and simulate future yield curves. Models must specify how the short-term rate evolves and how that short rate determines the rest of the term structure.

A basic continuous-time stochastic process

In a basic one-factor model, the short rate r_t is assumed to follow a **stochastic differential equation (SDE)** of the form

$$\mathrm{d}r_t = b\,\mathrm{d}t + \sigma\,\mathrm{d}z_t,$$

where:

- b is the **drift** term representing the expected direction of the change in the short rate.

- σ is the **volatility** term representing the magnitude of random shocks.
- z_t is a standard Wiener process (Brownian motion).

> Properties of $\mathrm{d}z_t$
>
> The term $\mathrm{d}z_t$ is the small random shock over the short time interval $\mathrm{d}t$. It has mean zero and variance $\mathrm{d}t$:
>
> $$\mathbb{E}[\mathrm{d}z_t] = 0, \qquad \mathrm{Var}(\mathrm{d}z_t) = \mathrm{d}t.$$
>
> For intuition, over a small time step Δt, we often approximate it as:
>
> $$\Delta z_t \approx \sqrt{\Delta t}\,\varepsilon, \qquad \varepsilon \sim N(0,1).$$
>
> So random shocks scale with $\sqrt{\Delta t}$, not Δt.

The drift and volatility are constants in this simplest model. If $b = 0$, the expected change in r is zero and the variance of the change over a time interval of length T is $\sigma^2 T$. Models of this type are sometimes called **normal models** because rate changes are independent of the current level.

Itô process

In reality, drift and volatility are not constant; they can depend on the current level of the short rate r and time t. An **Itô process** allows this by writing

$$\mathrm{d}r_t = b(r_t, t)\,\mathrm{d}t + \sigma(r_t, t)\,\mathrm{d}z_t.$$

Here, $b(r_t, t)$ and $\sigma(r_t, t)$ can be functions of r_t and t. Itô processes are the building block of most one-factor term-structure models.

Specifying the dynamics of the drift term

A common specification for the drift is **mean reversion**. The idea is that interest rates tend to revert toward a long-run average $\bar{r}$. The **speed of mean reversion** is measured by a parameter $\alpha > 0$. The drift becomes

$$b(r_t, t) = -\alpha(r_t - \bar{r}).$$

When r_t is above the long-run mean, the drift is negative; when below, it is positive. The speed α controls how quickly r_t moves back toward $\bar{r}$.

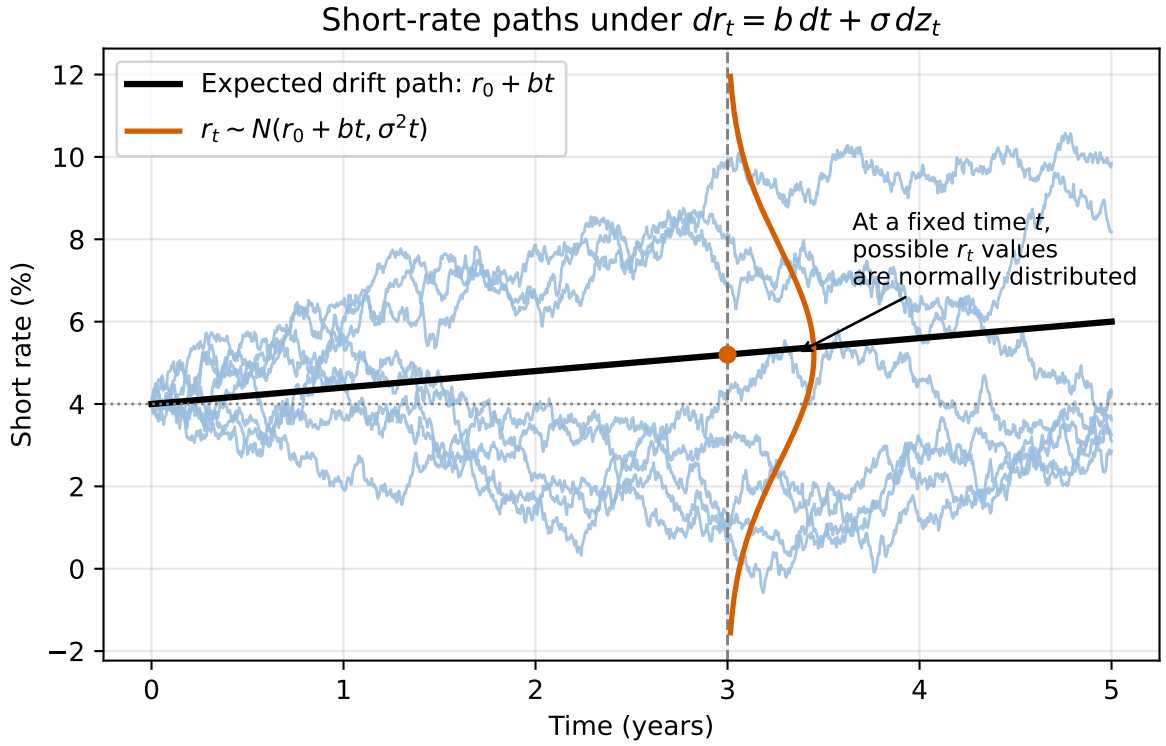


Figure 1: Drift gives the expected path; at each future time, the short rate is normally distributed around that path



Figure 2: Low versus high speed of mean reversion

Specifying the dynamics of the volatility term

Volatility can depend on the level of rates. A flexible specification is the **constant-elasticity-of-variance (CEV)** model, which sets

$$\sigma(r_t, t) = \sigma_0 \, r_t^{\gamma},$$

where $\sigma_0 > 0$ is a scale parameter and γ controls how volatility depends on r_t. Special cases include:

- $\gamma = 0$ (**Vasicek model**): volatility is constant and independent of r_t. This is a **normal model**.
- $\gamma = 1$ (**Dothan model**): volatility is proportional to r_t. This is a **lognormal model** (changes are proportional to the level).
- $\gamma = \frac{1}{2}$ (**Cox–Ingersoll–Ross model**): volatility is proportional to $\sqrt{r_t}$. Under the model's standard parameter restrictions, rates remain nonnegative because $\sigma_0 r_t^{1/2}$ goes to zero as r_t approaches zero.

Combining mean reversion in drift with square-root volatility yields the **mean-reverting square-root** model:

$$\mathrm{d}r_t = -\alpha(r_t - \bar{r})\,\mathrm{d}t + \sigma_0 \sqrt{r_t}\,\mathrm{d}z_t.$$

Example: Simulating an Ornstein–Uhlenbeck process

An Ornstein–Uhlenbeck (Vasicek) process has constant volatility and mean-reverting drift. The plot shows simulated paths for the following set of parameters:

- speed of mean reversion: $\alpha = 0.5$
- long-run mean: $\bar{r} = 0.05$
- volatility: $\sigma_0 = 0.02$
- starting short rate: $r_0 = 0.04$
- time horizon: $T = 5$ years
- time step: $\Delta t = 1/252$ years (one trading day)
- number of simulated paths: 5

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3 Arbitrage-Free versus Equilibrium Models

Interest-rate models can be classified by how they determine the term-structure evolution.

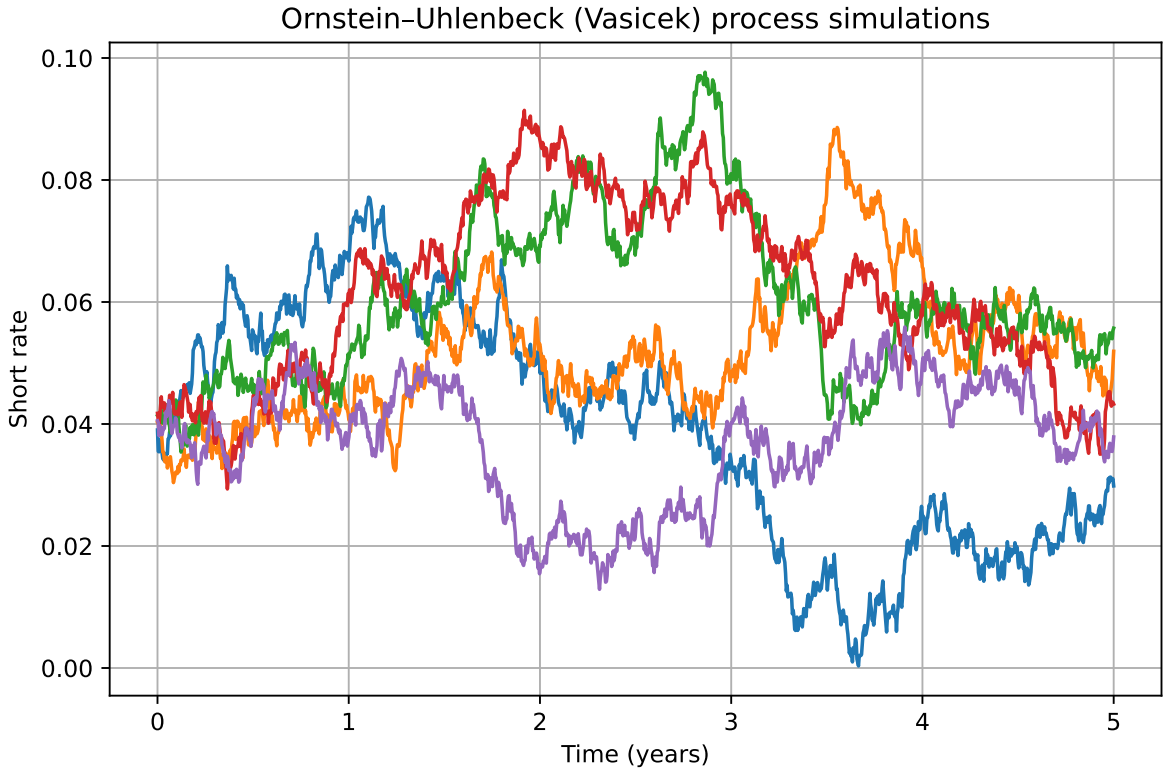


Figure 3: Simulated short-rate paths following an Ornstein–Uhlenbeck process

Arbitrage-free models

Arbitrage-free or **no-arbitrage models** start with observed market prices of a set of benchmark instruments (Treasury bonds, swaps, etc.). The model assumes a random process for the short rate with unknown drift. Parameters are calibrated so that the model exactly reproduces the current term structure and prices of benchmark instruments. Once calibrated, the model can value other instruments consistently.

Common arbitrage-free models include:

Model	Type	Key features
Ho–Lee	Normal	Constant volatility; no mean reversion. Produces normally distributed rate changes.
Hull–White	Normal	Adds mean reversion to the Ho–Lee model. Useful for pricing callable bonds.
Kalotay–Williams–Fabozzi (KWF)	Lognormal	Models the logarithm of the short rate; no mean reversion.
Black–Karasinski	Lognormal	Lognormal model with mean reversion.
Black–Derman–Toy	Lognormal	Lognormal with endogenous mean reversion; calibrates to caps and floors.
Heath–Jarrow–Morton (HJM)	General	Multi-factor model specifying the volatility structure of forward rates. Many models (Ho–Lee, Hull–White, KWF, etc.) are special cases.

Because these models match observed prices, arbitrage opportunities are eliminated by construction. They are widely used in practice for pricing complex derivatives and bonds with embedded options.

Equilibrium models

Equilibrium models explain the term structure by appealing to economic fundamentals and investor preferences. They assume a particular utility function and derive the stochastic process for rates from equilibrium conditions. Parameters are estimated from historical data rather than calibrated to market prices. Examples include the **Vasicek** and **Cox–Ingersoll–Ross (CIR)** equilibrium models, as well as the **Brennan–Schwartz** and **Longstaff–Schwartz** two-factor models.

Equilibrium models provide economic intuition but are less commonly used to price derivatives because they do not necessarily fit the current term structure. In practice, arbitrage-free models dominate due to their ability to match observed prices and deliver consistent valuations.

4 When Do We Need a No-Arbitrage Short-Rate Model?

The first rule is simple: **do not use a stochastic interest-rate model unless the cash flows require one**.

For a plain-vanilla Treasury bond or a fixed-rate corporate bond, the promised cash flows are usually fixed. Once we have a discount curve, valuation is mainly:

$$P = \sum_i CF_i \times P(0, t_i),$$

where CF_i is the cash flow at date t_i and $P(0, t_i)$ is today's price of a zero-coupon bond that pays \$1 at t_i. Equivalently, $P(0, t_i)$ is the discount factor from t_i back to today. The discount factors can be bootstrapped from Treasury prices, swap rates, or other benchmark instruments. No model of how future interest rates might move is required.

If we calibrated a Hull–White model to the same initial discount curve, it would reproduce the same time-0 value for this plain bond. That is useful as a consistency check, but it does not add much for valuing the plain bond itself.

Short-rate models become important when the value of the security depends on **future interest rates**, not just today's discount curve. This happens when cash flows are option-like, path dependent, or reset based on future market rates.

Examples include:

- callable bonds;
- putable bonds;
- mortgage-backed securities;
- caps and floors;
- swaptions;
- structured notes;
- other bonds with embedded options.

> What to remember
>
> A yield curve is a snapshot of today's market. A short-rate model is a rule for how interest rates may evolve through time. Plain bonds usually need the snapshot. Embedded options and interest-rate derivatives need the evolution.

4.1 Yield curve versus short-rate model

The **yield curve** gives today's market-implied discount rates across maturities. It answers: "What discount factors are implied by market prices today?"

A **short-rate model** specifies the stochastic evolution of the instantaneous short rate r_t. It answers: "How might interest rates move in the future, and how would those future paths affect cash flows and discounting?"

The distinction matters because a bond with fixed cash flows can be valued directly from today's curve. A callable bond cannot. If rates fall, the issuer is more likely to call the bond. If rates rise, the bond is less likely to be called. The cash flows themselves depend on future rate paths, so a model of rate dynamics is needed.

Feature	Plain fixed-rate bond	Callable bond or rate derivative
Cash flows	Fixed or mostly fixed	State-dependent cash flows
Main valuation input	Today's bootstrapped discount curve	Future yield curves generated by a model
Typical method	Discounted cash flow	Backward induction, lattice, or simulation
Exercise decision	None	Compare continuation value with exercise value
Examples	Treasury bond, fixed-rate corporate bond	Callable bond, MBS, cap, floor, swaption

4.2 From a short-rate process to a yield curve

A one-factor short-rate model starts by specifying the evolution of r_t. Once we simulate or solve for possible paths of r_t, each path implies a discount factor:

$$\exp\left(-\int_0^T r_s\, ds\right).$$

Under risk-neutral pricing, the zero-coupon bond price is the expected discounted payoff:

$$P(0,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^T r_s\, ds\right)\right].$$

The continuously compounded zero rate follows from:

$$y(0,T) = -\frac{\log P(0,T)}{T}.$$

So the model generates the term structure through this chain:

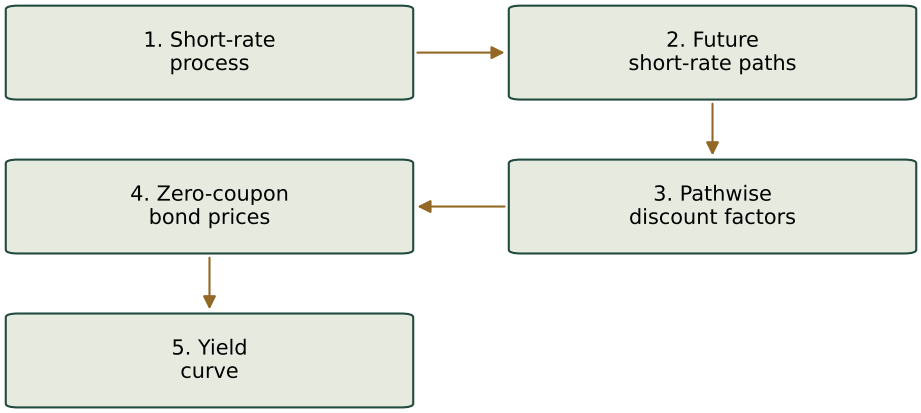


Figure 4: How a short-rate model generates the term structure

For time-0 pricing, $P(0,T)$ is observed from the market or calibrated to the market. For a future date $t > 0$, $P(t,T)$ is different: it is the model-generated price at time t of a zero-coupon bond maturing at T, with $T > t$.

In a one-factor Hull–White model, once a future short rate r_t is reached in the tree or simulation, that state determines the entire future yield curve at that date. In other words, each future node gives a full set of discount factors:

$$P(t,T) \quad \text{for all maturities } T > t.$$

4.3 A simple short-rate tree intuition

A lattice or tree is a discrete version of the same idea. Starting from today's short rate r_0, rates branch into possible future values. Each path has its own sequence of short rates and therefore its own discount factor. At each future node, the short rate implies a full future yield curve. For an option-embedded bond, the model also checks whether exercise is optimal at each node.

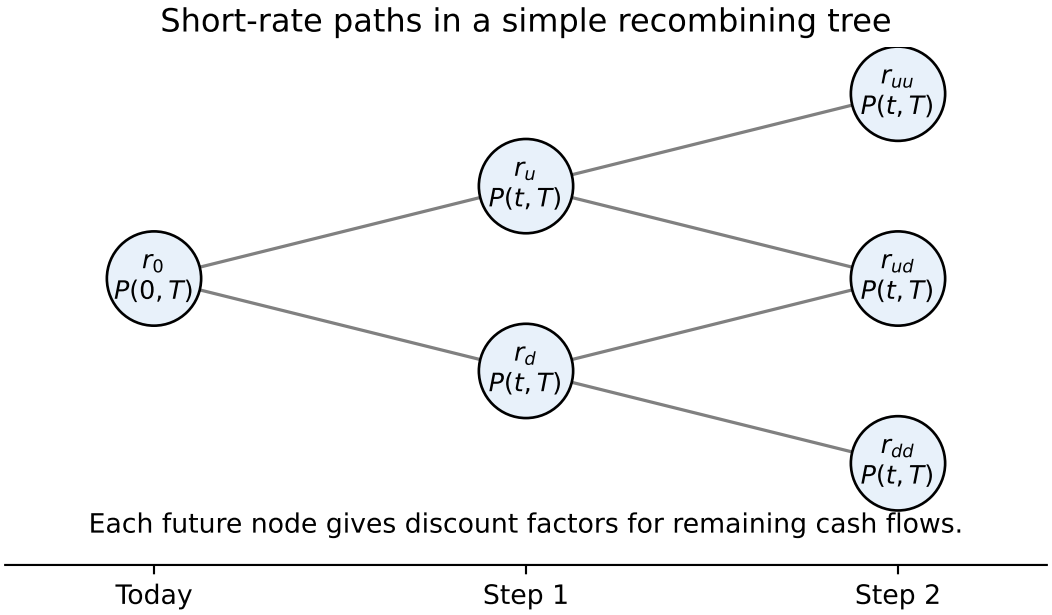


Figure 5: A simple short-rate tree: each future node implies a future yield curve

4.4 The Hull–White model

The one-factor Hull–White model is a no-arbitrage extension of the Vasicek model:

$$dr_t = [\theta(t) - ar_t]\,dt + \sigma\,dW_t.$$

Here:

- a is the speed of mean reversion;
- σ is the short-rate volatility;
- $\theta(t)$ is a time-dependent drift function;
- W_t is Brownian motion under the risk-neutral measure.

At first, the model may look like it has only two parameters, a and σ. If that were true, it could not match an entire observed yield curve. The key is that $\theta(t)$ is **not a single scalar parameter**. It is a function of time.

Hull–White uses $\theta(t)$ to fit today's initial term structure. The scalar parameters a and σ mainly control the future dynamics: how quickly rates mean-revert and how volatile they are. In practice, a and σ are often calibrated to interest-rate option prices such as swaptions.

This is the main value added by Hull–White. It is not mainly a tool for repricing today's plain fixed-rate bonds. After calibration, those prices already come from the observed discount

curve. Hull–White adds a no-arbitrage way to generate future short-rate scenarios and future yield curves that remain consistent with today's market.

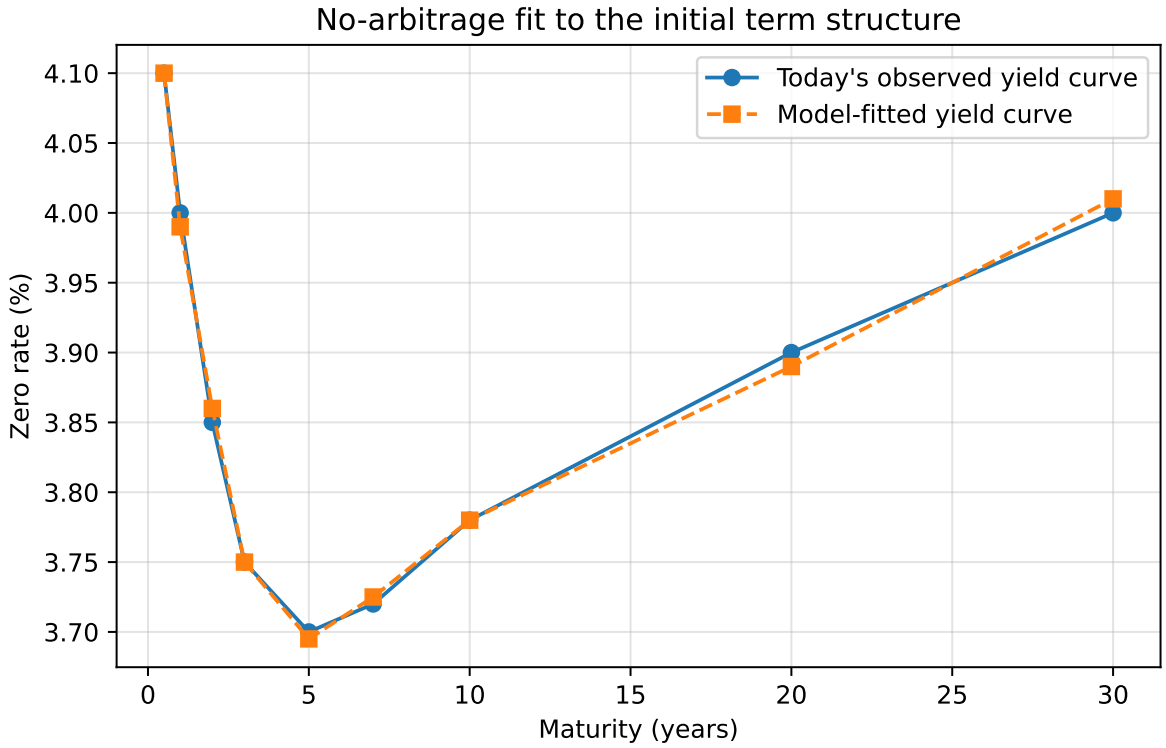


Figure 6: Hull–White calibration: theta(t) is chosen so the model fits today's observed curve

4.5 Calibration logic

The calibration workflow is usually:

1. **Bootstrap the market discount curve.** Use observed benchmark instruments such as Treasuries, deposits, futures, or swaps to infer market discount factors.
2. **Fit the initial term structure.** Choose $\theta(t)$ so that the model's zero-coupon bond prices match market zero-coupon bond prices.
3. **Fit future dynamics if needed.** Calibrate a and σ to liquid interest-rate options, such as caps, floors, or swaptions.
4. **Price the target instrument.** Use the calibrated model in a lattice, tree, finite-difference method, or Monte Carlo simulation.

For a plain fixed-rate bond, step 1 is usually enough. Steps 2–4 become valuable when the instrument's payoff depends on future rates or exercise behavior.

The full pricing logic for an option-embedded bond is:

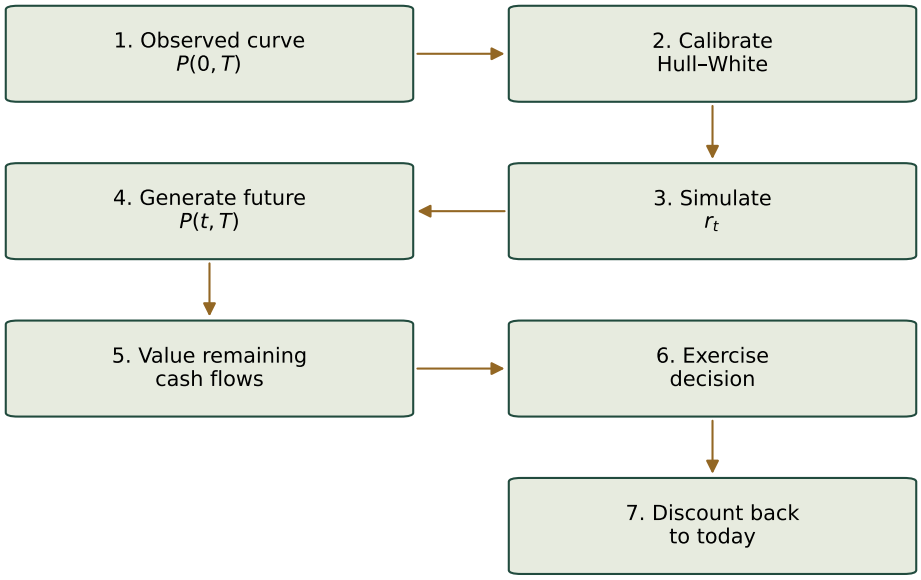


Figure 7: Hull–White pricing workflow for option-embedded bonds

Where the complexity enters

Without derivatives or embedded options, individual bond valuation is relatively straightforward. Fixed-income complexity usually comes from cash flows that change with future rates, embedded optionality, prepayment behavior, portfolio construction and risk management, and credit, liquidity, or spread modeling.

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5 Empirical Evidence on Interest-Rate Changes

When choosing a model, it helps to understand actual interest-rate behaviour. Two issues are often discussed: the relationship between volatility and the level of rates, and the possibility of negative rates.

Volatility of rates and the level of interest rates

Empirical studies of Treasury yield curves from the late 1970s to the 2000s show that the correlation between interest-rate volatility and the level of rates changes across regimes. During periods when rates exceeded roughly 10% (late 1970s and early 1980s), volatility tended to increase with the level of rates. When rates were below 10%, the correlation weakened and volatility behaved more like a constant. This evidence suggests that in low-rate environments (common since the 1990s), the **normal model** (constant volatility) may be adequate, whereas in high-rate regimes a lognormal or square-root model might better reflect the link between level and volatility.

Negative interest rates

Historical episodes show that nominal rates can occasionally be negative. During the U.S. Great Depression, Treasury bills traded at negative yields; more recently some short-term Japanese yen deposits briefly had negative yields in the late 1990s. Models with constant volatility allow negative rates, while models where volatility is proportional to the rate (lognormal) do not. The probability of negative rates and their impact on derivative values depend on the initial curve, volatility, horizon, and market regime. Normal models allow negative rates; that feature must be assessed against the application rather than dismissed as negligible.

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6 Selecting an Interest-Rate Model

When choosing a model for bond portfolio management, consider:

1. **Purpose of the model.** For pricing complex derivatives or option-embedded bonds, use arbitrage-free models that fit the current term structure. For economic insight or macroeconomic forecasting, equilibrium models may suffice.
2. **Volatility assumptions.** In low-rate environments, constant volatility models (Vasicek, Ho–Lee, Hull–White) perform well. In high-rate regimes or when negative rates must be ruled out, lognormal or square-root volatility (CIR, Black–Karasinski) may be preferred.
3. **Computational tractability.** Models should allow efficient calibration and pricing. Multi-factor models offer flexibility but are more difficult to implement.
4. **Data availability.** Arbitrage-free models require reliable market prices for a reference set of instruments; equilibrium models require long historical series to estimate parameters.

Ultimately, practitioners often choose the simplest model that fits the current term structure and provides robust valuations across scenarios.

7 Estimating Interest-Rate Volatility Using Historical Data

Volatility is an important input for any rate model. Two common methods are **historical volatility** and **implied volatility**.

Historical volatility

Historical volatility uses past interest-rate observations to estimate the standard deviation of rate changes. Suppose r_t is a series of weekly interest rates and Δr_t is the change from one week to the next. The sample standard deviation of Δr_t provides a weekly volatility. To annualize, multiply by $\sqrt{52}$ (assuming 52 weeks per year). For daily data, multiply the daily standard deviation by $\sqrt{n}$, where n is the number of trading days in a year (commonly 250 or 260).

Numerical example (artificial data)

Consider a hypothetical sequence of weekly short rates (in percent) over 10 weeks:

Week	Rate (%)	Weekly change (Δr_t, bps)
1	2.00	–
2	2.05	+5
3	2.04	−1
4	2.08	+4
5	2.02	−6
6	2.10	+8
7	2.09	−1
8	2.12	+3
9	2.15	+3
10	2.14	−1

To compute the sample standard deviation of the changes, convert basis points to decimal form (e.g., +5 bps = 0.0005). Using the nine non-missing changes (+5, −1, +4, −6, +8, −1, +3, +3, −1) we find:

1. Mean change = $\frac{5-1+4-6+8-1+3+3-1}{9} = \frac{14}{9} \approx 1.556$ bps.
2. Variance = $\frac{1}{8}\sum_{i=1}^{9}(\Delta r_i - \text{mean})^2$. Calculating each squared deviation and summing yields ≈ 140.22 (bp)^2. The sample variance is $140.22/8 \approx 17.53$ (bp)^2.

3. Standard deviation = $\sqrt{17.53} \approx 4.19$ bps per week.
4. Annualized volatility = $4.19 \times \sqrt{52} \approx 30.19$ bps per year.

Thus, this simplified data set implies an annual volatility of about 30.19 basis points.

Implied volatility

Implied volatility derives from market prices of option-like instruments (caps, floors, swaptions). An option pricing model expresses the option price as a function of volatility. By inverting the model using observed prices, one obtains a model-dependent, risk-neutral volatility input. Implied volatility reflects current market conditions and forward-looking expectations, whereas historical volatility reflects past behaviour.

Key Points

- An interest-rate model specifies a stochastic process for the short rate and derives other rates from it. Drift, volatility, and mean reversion are key components.
- One-factor models use a single source of uncertainty (Brownian motion). Mean-reverting drift and level-dependent volatility are common specifications.
- Arbitrage-free models (Ho–Lee, Hull–White, KWF, Black–Karasinski, BDT, HJM) calibrate to market prices and are widely used for valuation. Equilibrium models (Vasicek, CIR, Brennan–Schwartz, Longstaff–Schwartz) derive rates from economic assumptions.
- Plain fixed-rate bonds usually do not need a stochastic short-rate model; a bootstrapped discount curve is typically enough.
- Short-rate models are central when cash flows depend on future rates, such as callable bonds, mortgage-backed securities, caps, floors and swaptions.
- In the Hull–White model, $\theta(t)$ is a time-dependent function calibrated to fit today's term structure; a and σ control future mean reversion and volatility.
- $P(0,T)$ is today's observed or calibrated discount factor; $P(t,T)$ for $t > 0$ is a model-generated future discount factor used to value remaining cash flows.
- Callable bond valuation requires future yield curves because the exercise decision compares the continuation value with the call price.
- Empirical evidence suggests that volatility is largely independent of the level of rates when rates are below about 10%, supporting the normal model; at higher levels, volatility may increase with rates, favouring lognormal specifications.
- Normal models admit negative rates. Their probability and valuation impact depend on parameters, horizon, and regime.
- Historical volatility is computed from past rate changes and annualized using square-root-of-time scaling; implied volatility is extracted from option prices.

Suggested Readings

Fabozzi, *Bond Markets: Analysis and Strategies*, Chapter 17.

Real-Market Data and Resources

- U.S. Treasury, Daily Treasury Par Yield Curve Rates: Observe the market term structure that arbitrage-free short-rate models are calibrated to reproduce.
- Federal Reserve Bank of New York, Reference Rates: Access official SOFR and related overnight funding-rate data used to anchor the front end of U.S. interest-rate models.
- FRED, Effective Federal Funds Rate: Download a long daily history of the policy-sensitive overnight rate for studying short-rate dynamics, volatility, and mean reversion.
- Federal Reserve, Selected Interest Rates (H.15): Compare current and historical Treasury, policy, and private short-term rates in an official weekly release.